TwitterFeel

Study progression and findings

Research record | Edition 1 | Compiled 2026-10-21

This record follows the study from its original forecasting question to the current existing-post emotion and sentiment workbench. It brings together findings, limitations, and changes in interpretation that were spread across the repository. It is a retrospective synthesis of saved evidence, not a new experiment or an independently reviewed publication.

How to read the dates and evidence

Dates are shown in America/New_York where a timestamp is available. A recorded date comes from an explicit experiment entry, report, or artifact timestamp. An archive date comes from Git and does not establish when an experiment ran. Undated original work is placed first for context, without assigning it a year. E1-E12 were reconstructed on 2026-07-17; their execution dates are unknown. All entries in this edition were backfilled on 2026-10-21.

Results marked **reported** come from saved documents or artifacts; they were not rerun for this record. Checks marked **reproduced** were performed during the 2026-10-21 review. A **hypothesis** or **planned** item remains unresolved. Later corrections are dated explicitly rather than silently changing the past.

Progression at a glance

Date or date basis	Change in the study
Original work, undated	Proposal moves from mental-health risk to next-tweet emotion prediction
2026-05-26, Git archive	Sustained sadness-label pattern analysis is preserved
2026-07-17, reconstructed	E1-E12 explore text, time, task scope and model architecture
2026-07-17 and 2026-07-28	More users and different class weighting yield limited gains
2026-09-22	V4 introduces stronger baselines and auditable grouped evaluation
2026-10-15	Existing-post labeling, public references, thresholds and model comparison
2026-10-16	Execution checks expose concrete labeling weaknesses
2026-10-21	Historical audit confirms saved-code defects and evaluation uncertainty

Scores from different tasks are not directly comparable. Six-class forecasting accuracy, binary forecasting accuracy, and multi-label micro/macro F1 have different meanings. Source IDs such as [S01] refer to the final source register.

Original study | Date not established

Evidence: historical documents; reported.

Question and method changed during the original project

The proposal asks whether social-media patterns can predict future mental-health conditions. Its proposed method is TF-IDF with XGBoost on four categories of existing tweets. Such a classification experiment alone would not evaluate future onset of a condition. The final report instead asks how to predict the emotion of a user's next tweet using their tweet history. [S01, S02]

The final workflow describes roughly 20,000 tweets, six transformer-generated emotion labels, weekend and overnight features, batches of ten posts from one user, an LSTM, and SMOTE. The saved preprocessing uses the first nine posts as history and the tenth emotion as the target. Users with fewer than 20 or more than 1,000 posts are excluded in the filtered variation. [S02, S03]

Finding as originally reported

The filtered-data/SMOTE variation reports approximately 0.82 accuracy, precision, recall and F1. A later audit, dated 2026-10-21, recalculates its matrix as 0.8233 accuracy and 0.8182 weighted F1. The original report acknowledges that oversampling increased prediction counts, but does not establish the split or resampling order. The source implementation of that training run is absent from the inspected history. These are reported results with unresolved evaluation provenance. [S02, S03]

What this changed

Subsequent repository experiments pursued reproduction of the high forecasting score. The research target was a classifier-generated emotion label, rather than an independently observed emotional state or clinical outcome. [S04, S05]

2026-05-26 | Sustained sadness-label patterns

Date basis: Git archive, commit e3ca7a0; execution date unknown. Reported.

An archived side study examines rolling sadness-label frequencies relative to each user's full-history baseline. Its saved report analyzes 72 users and records 5 users with at least one flagged period, totaling 24 periods. Settings include 20-post windows, a 0.40 absolute threshold, a 0.15 excess over baseline, and at least three consecutive flagged windows. [S06]

These counts describe behavior of a text classifier and a selected threshold rule. They do not validate depression detection. The use of a full-history baseline also makes this a retrospective analysis; a prospective version would need a baseline calculated only from information available at each time. That prospective restriction is an interpretation added in this edition.

2026-07-17 | Forecasting reproduction, E1-E12

Date basis: reconstructed from early notes on this date. Execution dates unknown. Results reported; qualifications below were added 2026-10-21.

The experiments progressively add text embeddings, sliding windows, richer time features, larger histories, different architectures and simpler prediction tasks. User-disjoint splits aim to measure transfer to unseen users. [S04]

Six-class and four-class variants

Experiment	Task and main change	Accuracy	Weighted F1
E1	Six classes; emotion history and time flags	0.232	0.279
E2	Six classes; MiniLM text and sliding windows	0.364	0.342
E5	Six classes; richer time features, window 10	0.389	0.379
E6	Six classes; window 16	0.415	0.368
E11	Four classes; omit love/surprise, window 10	0.457	0.410
E12	Four classes; window 16	0.439	0.399

E1's saved majority and persistence accuracies are 0.348 and 0.410 on 293 test windows. E2's majority accuracy is 0.405 on a different window set. E11 exceeds its majority baseline of 0.416, but changes the task by excluding two target classes. Its macro F1 is only 0.247. Improvements in headline accuracy do not establish reliable performance across emotions. [S04, S07]

Binary variants

Experiment	Main change	Accuracy	Weighted F1
E3	Text embeddings, user-disjoint split	0.570	0.559
E4	Row-level random split	0.595	0.595
E7	Richer time features, window 10	0.578	0.553
E8	BiLSTM and attention, window 10	0.571	0.560
E9	LSTM, window 16	0.562	0.559
E10	BiLSTM and attention, window 16	0.553	0.553

E3 approximately ties its 0.568 majority baseline. Larger windows and attention do not consistently improve these configurations. E4 allows the same users across partitions, but also changes class weighting and test composition; its roughly 2.5-point difference from E3 does not isolate the effect of leakage. Neither binary comparison explains the original six-class SMOTE result. [S04]

Progression: text helps relative to E1, but architecture changes do not recover the original score. The later runs use different encodings, training choices and evaluation partitions from the original report. The original result remains unreproduced by these saved experiments.

2026-07-17 | More users: Sentiment140, E13

Date basis: dated experiment entry. Reported.

A corpus swap retains 1,405 users and 50,654 sampled tweets, compared with roughly 38 training users in the earlier corpus. The same general labeler, embedding and V3 forecasting workflow is used. [S04]

Task	Accuracy	Weighted F1	Majority accuracy	Persistence accuracy
Six classes	0.215	0.254	0.501	0.391
Four classes	0.299	0.334	0.520	0.404
Binary	0.572	0.563	0.538	0.548

The binary run exceeds both baselines, with ROC-AUC 0.593. Multi-class results remain weak. The log proposes that strong class weighting under severe imbalance is distorting the optimization objective. This is an explanation to test, rather than a controlled demonstration of the cause. Sentiment140 is an emoticon-search sample, so the next retained tweet is not necessarily the next tweet actually posted by a user. [S04]

2026-07-28 | Class-weight tradeoffs, E14

Date basis: dated experiment entry. Reported.

The follow-up changes class weights on the original corpus. The intended Sentiment140 follow-up remains unperformed in the saved record because its raw data/intermediates are unavailable. [S04]

Task	Weighting	Accuracy	Weighted F1	Macro F1
Six classes	Balanced	0.3887	0.3791	0.1725
Six classes	Square root	0.4612	0.4097	0.1738
Six classes	None	0.4338	0.3129	0.1259
Four classes	Balanced	0.4572	0.4099	0.2474
Four classes	Square root	0.4284	0.3036	0.1790
Four classes	None	0.4237	0.2679	0.1597

Six-class square-root weighting improves accuracy mainly through the dominant classes; rare-label performance remains poor. Four-class performance worsens when weights are softened. Class weighting is a task-specific tradeoff, and the six-class accuracy gain is not broad emotion recognition. [S04]

Reddit importer: implementation evidence only

A dated synthetic-fixture check covers cleaning, filtering, decompression and the embedding/window chain. No real Reddit corpus or resulting accuracy study is recorded. Labeler domain shift, subreddit shortcuts and UTC time features remain limitations; synthetic success does not resolve them. [S08]

2026-09-22 | V4: auditability and stronger baselines

Date basis: dated E15 entry and saved metrics. Reported. The entry was written after implementation, not preregistered.

V4 separates data auditing, grouped evaluation and model comparison. It removes 204 identical user/time/text records and skips 1,068 windows containing timestamp ties. The prepared data has 11,801 windows from 53 users. Five development folds use 45 users and 10,336 windows; eight reserved test users are not scored. [S04, S05, S09]

Binary development comparison

Model or baseline	Accuracy	Macro F1	Mean user accuracy
Training-majority	0.4709	0.4264	0.4637
Persistence	0.5600	0.5591	0.5489
Historical frequency	0.5833	0.5816	0.5846
Transition	0.5600	0.5591	0.5489
Logistic, text/time/emotion	0.5190	0.5188	0.5154
LSTM, text/time	0.5146	0.5142	0.4991
LSTM, text/time/emotion	0.5038	0.5036	0.4978

Historical frequency exceeds persistence by 2.33 percentage points. The paired user-cluster bootstrap interval is +1.09 to +3.77 points. Both LSTM variants trail the frequency baseline. Adding the existing historical emotion features does not help this particular LSTM configuration. [S09]

Data pipeline corrections

Sentiment140 spam ratios are now computed before deduplication; older outputs must be regenerated to benefit. Reddit decompression now streams through a temporary SQLite database rather than retaining all authors' text in memory. Temporary disk use and the eligible-author list still grow with the corpus, and the sampling budget can exceed its target by one author's history. Real Reddit data remains untested in this record. [S05]

Limits and changes in interpretation

- The run uses one-hot emotion history because the old CSV lacks soft labeler probabilities. This does not test newly generated probability features.
- Legacy embedding order was explicitly accepted without a verification manifest. V4 supports manifests for new artifacts, but that does not retrospectively verify this run's old embeddings.
- Labels are generated by a classifier. The target is the next retained post, and filtering can remove intervening posts.
- User-cluster intervals condition on fitted models; they omit refitting and broader model-selection uncertainty. The run is development evidence, not a newly scored final test result.
- UTC hour features do not establish a user's local overnight behavior. [S05, S09]

Progression: the useful outcome is a stronger comparison standard and a more auditable workflow. The evidence favors a simple history baseline over these learned configurations; it does not prove that forecasting is impossible or that larger architectures would solve the observed limitations.

2026-10-15 | Existing-post labeling and human references

Date basis: dated pilot artifact, public-data notes and v0.2.0 release. Reported. Order within this date is thematic unless timestamps establish it.

The product now imports existing text, suggests multiple emotions using a GoEmotions model, and runs a separate sentiment classifier. It saves dataset versions, model/settings provenance and observations. Applying these pretrained weights does not train them on imported posts or human reviews. [S10, S11]

Initial pilot: execution without human accuracy labels

The pilot processes 400 posts at threshold 0.5: 102 labeled, 235 neutral, 59 uncertain and 4 requiring review. Only four posts receive multiple non-neutral labels. Human-reviewed rows are zero and quality metrics are null. The status counts demonstrate output behavior, not the proportion correctly labeled. They also cannot establish improvement over the historical six-class labeler. [S12]

GoEmotions development reference

All 5,426 official development comments are evaluated. The 5,427 official test examples are not scored in this project evaluation. [S13, S14]

Shared threshold	Precision	Recall	Micro F1	Exact set match	Mixed full recovery
0.50	0.686	0.511	0.586	47.4%	9.4%
0.30	0.584	0.636	0.609	44.9%	19.5%

Lowering the cutoff retrieves more reference emotions and introduces additional incorrect labels. Mixed full recovery allows extra labels, while exact set match rejects them. Neither coverage nor recovery alone establishes quality.

The model was trained on GoEmotions; these results are an in-domain reference comparison, not independent Twitter/upload validation. The reference retains 174 neutral-plus-emotion development rows and repeated texts. Ten text hashes overlap the official development and test splits. The project's human rubric instead requires neutral alone, and the model author's documented test-based threshold tuning limits the test split's independence. [S13, S14]

Twitter-specific comparison remains incomplete

The SemEval adapter preserves its native labels and eight exact-name shared categories. The documented official download attempts returned HTTP 406; no actual reference preparation or model score was completed. Synthetic adapter tests do not fill that evidence gap. This is the recorded status on this date, not a fresh availability check. [S13]

2026-10-15 | Threshold refinement exposes tradeoffs

Date basis: artifact timestamp 2026-10-15T23:01:15Z, October 15 locally. Reported development research.

Five decision rules use the same 5,426 development rows and 5,423 exact-text groups. Learned cutoffs are fitted on four folds and applied to the remaining fold; the pretrained model is not retrained. There are 712 reference posts with multiple non-neutral emotions. [S15, S16]

Rule	Micro F1	Macro F1	Mixed full recovery
Fixed 0.50	0.585922	0.447795	9.41%
Historical 0.30	0.609231	0.496341	19.52%
Fine shared cutoff	0.607499	0.494251	18.82%
Joint micro per-label	0.614223	0.480234	17.70%
Per-label F1	0.605527	0.492843	19.80%

Finding

The joint-micro profile improves micro F1 over historical 0.30, but reduces macro F1 and mixed-emotion recovery. False-positive decisions fall from 2,886 to 2,613 while missed labels increase from 2,321 to 2,394. Seventeen labels lose F1, seven improve and four are unchanged. The highest aggregate score does not identify a universally better label-selection rule. [S15]

The paired micro-F1 difference interval is +0.000685 to +0.010064, from 500 exact-text-group resamples. It holds fitted decisions fixed and omits threshold fitting, method selection, annotation and upstream-model uncertainty. The historical 0.30 comparator was previously chosen on the full development set; this is not a comparison with an untouched baseline. [S15, S16]

Decision and remaining limits

No threshold candidate improves micro F1 while preserving both macro F1 and mixed-label recall against the historical comparator. Manual shared thresholds remain the default; exported profiles require explicit selection and matching model provenance. They change the label decision rule, not model weights or probability calibration. [S15]

Rare labels have inadequate fitting support, some fold cutoffs are unstable, and exact-text grouping cannot account for different comments by the same author. An objective-specific improvement on these development annotations still needs independent review on the user's target posts.

2026-10-15 | A different pretrained model

Date basis: saved run identifier beginning 20261015T230413Z, October 15 locally. Reported; exploratory comparison.

The study compares the existing GoEmotions RoBERTa model with one pinned ModernBERT candidate. Both receive the same 5,426 posts, references and grouped folds, using a 256-token limit. Different tokenizers can retain different amounts of text within that limit. [S15, S17]

Model and rule	Micro F1	Macro F1	Exact set match	Mixed full recovery
RoBERTa, shared 0.30	0.6092	0.4963	44.90%	19.52%
ModernBERT, shared 0.30	0.6195	0.5378	43.86%	24.44%
ModernBERT, fitted shared	0.6192	0.5358	44.95%	22.05%
ModernBERT, joint micro per-label	0.6211	0.5247	46.66%	18.82%

Finding

At shared 0.30, ModernBERT improves micro F1, macro F1 and mixed full recovery. Precision falls from 0.5844 to 0.5796, exact match declines, and neutral/emotion conflict flags rise from 736 to 946. Its highest-micro-F1 profile sacrifices some of the mixed-emotion benefit. [S15, S17]

The shared-0.30 paired micro-F1 difference is about +0.01030, with a conditional 95% text-group bootstrap interval of +0.00294 to +0.01708. This interval does not include candidate selection or upstream-model uncertainty. [S15]

What this changed

ModernBERT becomes an explicit research choice, with model-specific threshold profiles. The app retains RoBERTa as its default. Model choices, copied thresholds and file fingerprints are recorded with new analyses. [S11, S15]

Both models were trained on GoEmotions. The candidate was selected after earlier results, the model card identifies it as superseded in the saved record, and this bounded comparison does not survey all available models. No official test examples are scored here. The result supports a further target-data comparison, not a claim that the app is now more accurate on the user's posts. [S17]

Progression: changing the pretrained model yields a more useful mixed-label development comparison than optimizing one score through thresholds alone. Whether that benefit transfers to Twitter or the user's uploads is unresolved.

2026-10-16 | Execution parity and concrete error cases

Date basis: dated diagnostic notes, changelog and handoff. Reported.

Twelve synthetic scenarios exercise the CLI and native desktop inference. The eleven nonblank inputs agree within a 0.00001 score tolerance, with a maximum absolute difference around 0.0000000103. Labels and statuses match; blank text is invalid or rejected. This demonstrates execution consistency on the constructed examples. It is not an accuracy benchmark. [S18]

Labeling weaknesses observed at threshold 0.50

Scenario	Observed result	Limitation exposed
Excited to move, terrified to leave	Fear only	Mixed emotions can be missed
Happy about promotion, sad about departure	Sadness only	Multi-label support does not guarantee full recovery
Sarcastic cancelled-train complaint	Admiration	Literal positive wording can mislead the model
Context-dependent ambiguous statement	Neutral	Missing context can be mistaken for neutrality
Spanish mixed-emotion sentence	Uncertain	One abstention does not validate language detection
Long repeated text	Truncated; review needed	Only retained tokens contribute to prediction

These judgments are provisional project interpretations. There are no independent human gold labels or adjudication for these cases. They identify challenge categories to study, not measured population error rates. [S18]

Runtime checks have a different scope

The handoff records 108 passing unit tests, a passing JavaScript syntax check, and a native smoke test with no JavaScript errors or listening TCP socket. The smoke test exercises application flows with disposable research state. The duplicate browser UI is retired and the desktop paths share an inference engine. These checks support implementation behavior; they do not establish label accuracy or scientific validity. [S10, S19]

Human-review evidence remains to be built

The recommended workflow uses a 30-post blind practice round to learn the rubric, then a separate representative target-data benchmark with a distinct challenge set. The practice round alone cannot establish reliability. Reviewers' judgments stay separate from model predictions. Agreement, disagreements, adjudication, rare-label support and abstentions need to be reported. [S20]

Some older Markdown notes describe per-label thresholds as future work; the October 15 artifacts document their later implementation. That completion does not complete independent human review or establish target-data accuracy.

2026-10-21 | Historical audit and corrections

Date basis: review conducted on this date. Arithmetic and saved-code defects reproduced; original training procedure remains unknown.

The original report, preserved notebook and repository history are compared directly. The page-6 confusion matrix yields 694 correct predictions out of 843, accuracy 0.8232503 and weighted F1 0.8182259. The approximate 0.82 claim has supporting arithmetic. [S02, S03]

Evaluation provenance is the larger uncertainty

The matrix's class supports are [157, 154, 143, 127, 118, 144]. Reconstructing V1 windows from the current saved corpus produces 1,323 complete windows with target counts [185, 607, 16, 408, 100, 7]. Under the preserved mapping, class 5 is surprise; the entire filtered corpus has only 64 surprise-labeled posts. If the original run used those same labels and windows, its 144 class-5 test examples cannot all be distinct untouched original windows. [S03]

This is strong reason to investigate evaluation resampling or different source artifacts. It does not prove the original order of SMOTE and splitting, or establish that leakage alone explains the score.

Two saved-code defects were reproduced

- The notebook declares 29 columns but assigns 28-value rows. A clean replay raises a mismatched-column error. The CLI preprocessing uses 28 columns.
- Incorrect final-batch slicing produces oversized trailing batches. For 21 posts, batch lengths are 10, 10 and 29; the row-length filter discards the last batch. Later versions use complete sliding windows. [S03]

All four notebook versions found in accessible history contain preprocessing only. They do not contain the original training, SMOTE or splitting code. Saved-code defects are confirmed; their role in the original score is unknown.

Corrections and local reproducibility

The E1 baseline summary is corrected to majority 0.348 and persistence 0.410 to match its saved metrics. The E4 explanation no longer attributes its whole gain to leakage because class weighting and test composition also changed. Prior conclusions must be read with these dated qualifications. [S03, S04]

The checkout has no outgoing commits at the start of this review; apparent Git changes were line-ending differences. The current machine lacks the EXE, workbench environment and ignored runs/model cache described in older notes. Legacy arrays exist at the root rather than current pipeline paths. No model was retrained and no full application test was rerun during this audit. [S03, S19]

2026-10-21 | Evidence gaps and the next study record

Synthesis as of this edition. Items below are open work, not findings from completed experiments.

Questions that remain open

- **Original score:** recover the actual training/SMOTE script, frozen labels, split indices and predictions. Check resampling order, untouched evaluation rows, shared users/windows, target-feature separation and categorical encoding.
- **Forecasting:** confirm old embedding alignment or regenerate verified artifacts; compare probability-history features and baselines on fixed development users before a final test. More architecture alone is not an established answer to the observed weak signal.
- **Existing-post quality:** collect independent human judgments on target posts, separating representative sampling from enriched challenge examples. Assess mixed emotions, sarcasm, ambiguity, unsupported language and truncation.
- **External comparison:** complete a documented Twitter-specific benchmark when the official data is accessible, preserving its label scope. Existing GoEmotions results do not fill this gap.
- **Reproducibility:** restore the local environment and provenance artifacts before claiming that historical runtime checks apply to this checkout.

Current interpretation

The study has progressed from chasing a high forecasting score toward more explicit targets, stronger baselines, preserved source identities and independent review requirements. The record contains useful development comparisons and reproducible software checks. Reliable forecasting at the original reported level, improved labeling on the user's posts, and clinical interpretation of emotion outputs have not been established by the saved evidence.

How to add the next entry

Keep this Markdown file as the editable master and regenerate the PDF. Append new dated entries before the source register. If an old conclusion changes, record a dated correction that points to the earlier entry. Do not overwrite old scores or assign an execution date from a file modification timestamp.

Each entry should state the event date and its basis; the question; task, dataset and split; what changed; results with baseline and sample size; the supported conclusion; limitations; next question; and source/run references. Mark proposals, synthetic tests, development results and final held-out results explicitly. Add new sources to the register and retain underlying artifacts.

Rebuild from the repository root with:

```
python -m pip install reportlab

python tools/build_study_progression.py
```

The builder reads this file, creates the PDF and a source-hash manifest under `output/pdf/`. Review the rendered pages after substantive edits.

Source register

Paths are relative to the repository unless marked external. This is a register of project evidence, not a fresh verification of external links or dataset availability. The build manifest records SHA-256 hashes of available sources. Missing ignored run artifacts limit reproduction even when summaries remain.

- **[S01]** External: `Project Proposal Presentation.pptx`, user-supplied from the local Downloads folder. Original proposal; date not established.
- **[S02]** External: `Social Media Project Report.pdf`, user-supplied from the local Downloads folder. Final original report, especially pages 3-8.
- **[S03]** `docs/HISTORICAL_REVIEW.md`; saved notebook `docs/CAP4773_fin_1.ipynb`; `src/preprocess.py`; `data/interim/dataset_godknowswhat.csv`. Review dated 2026-10-21.
- **[S04]** `docs/EXPERIMENTS.md`. E1-E15, including dated qualifications.
- **[S05]** `docs/V4.md`. Historical forecasting protocol and known limits.
- **[S06]** `twitter_feel_plus/results/REPORT.md`, `params.json`, `aggregate.json`; `twitter_feel_plus/README.md`; `sadness_flag.py`. Report first archived in commit `e3ca7a0` on 2026-05-26.
- **[S07]** `metrics/metrics.json`, `metrics_v2.json`, `metrics_binary.json`, `metrics_leaky_binary.json`, and the corresponding `metrics_windows_v3_*.json`. Full files are under `metrics/`; metric filenames abbreviated here only.
- **[S08]** `docs/REDDIT.md`; `src/prepare_reddit.py`. Synthetic verification dated 2026-07-28; later ingestion fixes documented in [S05].
- **[S09]** `metrics/metrics_v4_mht_binary_development.json`. Dated 2026-09-22.
- **[S10]** `CHANGELOG.md`; `app_version.json`. v0.2.0 and workspace cleanup.
- **[S11]** `docs/WORKBENCH.md`; `src/analysis_engine.py`; `src/analysis_jobs.py`. Current workflow and inference behavior.
- **[S12]** `metrics/emotion_labeling_pilot_v1.json`. Dated 2026-10-15.
- **[S13]** `docs/PUBLIC_DATASETS.md`. Availability checks dated 2026-10-15.
- **[S14]** `metrics/goemotions_development_v1.json`. Public-reference results.
- **[S15]** `docs/THRESHOLD_REFINEMENT.md`. Threshold/model tradeoffs and caveats.
- **[S16]** `metrics/emotion_threshold_refinement_v1.json`. Dated timestamp.
- **[S17]** `metrics/emotion_model_comparison_v1.json`. Timestamped run ID.
- **[S18]** `docs/LABELING_CHECK.md`. Diagnostic dated 2026-10-16.
- **[S19]** `HANDOFF.md`. Current scope, prior runtime checks and checkout audit.
- **[S20]** `docs/EMOTION_LABELING.md`. Annotation and evaluation protocol; some planning statements are superseded by later artifacts [S15-S17].

Edition history

Edition 1, 2026-10-21: backfilled the historical documents, forecasting log, sadness-pattern report, V4 comparison, labeling pilot, public-reference work, threshold/model comparisons, execution diagnostics and historical audit. No new model training or final held-out evaluation was performed for this edition.